SMART INDIA HACKATHON 2025

- **Problem Statement ID –25049**
- **Problem Statement Title- AI-Driven Public Health Chatbot for Disease Awareness**
- **Theme- MedTech / BioTech / HealthTech**
- **PS Category- Software/Hardware**
- **Team ID-**
- **Team Name (Registered on portal)- tech4graze**

TECH4
GRACE

GRACE HEALTH BOT

SMART INDIA HACKATHON 2025

Create a multilingual AI chatbot to educate rural and semi-urban populations about preventive healthcare, disease symptoms, and vaccination schedules. The chatbot should integrate with government health databases and provide real-time alerts for outbreaks.

We propose an **AI-powered multilingual health chatbot** tailored for rural and semi-urban populations to provide preventive healthcare education, symptom guidance, vaccination reminders, and real-time outbreak alerts. It supports multiple regional languages and voice interaction, accessible via mobile apps, SMS, and IVR. Offline capabilities ensure usability in areas with poor internet. Integration with government health databases ensures up-to-date, reliable information.

How It Addresses the Problem:

Overcomes language and literacy barriers through multilingual and voice-enabled AI.
Provides timely, personalized health advice and alerts via direct government data integration.
Offers offline access and multi-channel support to reach remote populations.
Empowers communities with knowledge, improving early disease detection and vaccination compliance.

Innovation and Uniqueness:

Combines advanced NLP with voice and offline functionality—rare in rural health tools.
Real-time integration with official health data ensures accuracy.
User-friendly for low-literacy populations through simple, conversational interfaces.

TECH4 GRACE

FEASIBILITY AND VIABILITY

SMART INDIA HACKATHON 2025

Feasibility Analysis

•**Technical Feasibility:** Available AI/NLP tools support multilingual and voice capabilities; government APIs enable reliable data integration.
•**Operational Feasibility:** Multichannel access (mobile app, SMS, IVR) ensures broad reach.
•**Economic Feasibility:** Cloud infrastructure and open-source AI frameworks reduce costs; scalability lowers per-user expense.
•**Social Feasibility:** Aligns with government health initiatives; high potential acceptance due to local language support.

Potential Challenges and Risks

•**Connectivity Issues:** Mitigated by offline functionality and SMS/IVR fallback.
•**Language Diversity:** Continuous content localization and AI model fine-tuning required.
•**Data Privacy & Security:** Must comply with health data regulations; implement encryption and anonymization.
•**User Literacy:** Voice-based interaction and simple UI design reduce barriers.
•**Integration Delays:** Close collaboration with government bodies to ensure API availability.

TECH4 GRACE

TECHNICAL APPROACH

SMART INDIA HACKATHON 2025

TECHNOLOGIES

Python, JavaScript (React Native), Java/Kotlin, Swift
NLP: Rasa NLU, Hugging Face Transformers
Voice: Google Speech API, Mozilla DeepSpeech
Backend: FastAPI, Node.js, PostgreSQL, MongoDB
Cloud: AWS/Azure/GCP
Integration: Government health APIs
Channels: Mobile apps, SMS, IVR

METHODOLOGY

- Requirement Analysis & Design
- Prototype Development
- AI & Multilingual NLP Integration
- Government API Integration
- Multi-channel Deployment (App, SMS, IVR)
- Pilot Testing & Iteration
- Full Deployment & Monitoring

TECH4 GRACE

IMPACT AND BENEFITS

SMART INDIA HACKATHON 2025

Impact

- Enhanced health literacy and self-care capabilities in rural communities.
- Early disease detection and prevention leading to fewer hospitalizations.
- Increased vaccination rates reducing infectious disease burden.
- Empowered community health workers with digital tools.

Benefits

- **Social:** Improved public health outcomes, reduced healthcare disparities.
- **Economic:** Lower healthcare costs due to prevention; increased productivity.
- **Environmental:** Reduced need for travel to healthcare centers lowers carbon footprint.
- **Scalability:** Can be adapted to other regions or countries with similar challenges.

TECH4 GRACE

RESEARCH AND REFERENCES

SMART INDIA HACKATHON 2025

References and Research Work Government Health API Documentation (e.g., India's Ministry of Health and Family Welfare) Research on AI chatbots for healthcare in low-resource settings (e.g., Journal of Medical Internet Research) NLP frameworks for regional languages (Hugging Face multilingual models, Rasa NLU) Case studies on digital health interventions in rural India (WHO reports, NITI Aayog publications)